

Statement of Work

Taste Matcher v3.1 Refactor

Project: Taste Matcher

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1. Project Objective

This Statement of Work governs the v3.1 refactor sprint of Taste Matcher, a local Node.js application that predicts movie ratings based on personal Letterboxd history. The primary objectives of this sprint are:

- Resolve critical data integrity flaws — specifically phantom duplicate entries in the ratings and watchlist state
- Consolidate the mathematical scoring model into a single, unified `scoreItem()` function used across all code paths
- Implement strict compliance with the Letterboxd list export v7 CSV standard
- Deliver a complete Software Engineering documentation package (SRS, SDS, Test Plan)

2. Scope of Work

2.1 Data Integrity

Implement triple-validation duplicate prevention for all manual rating and watchlist entry endpoints:

- Primary check: Letterboxd URL match via `normalizeUrl()`
- Secondary check: Title + Release Year match via `titleKey()`
- Tertiary check: TMDb ID match

Any match on any of the three identifiers returns HTTP 409 with a structured error payload. The add-rating endpoint additionally handles the `fromResolved` flag to allow legitimate rating updates from the unresolved-film flow.

2.2 Export Compliance

Rewrite the CSV generation module to produce a Letterboxd v7-compliant export:

- File begins with the literal string 'Letterboxd list export v7' on line 1
- List metadata row (Date, Name, Tags, URL, Description) follows on line 2
- Film rows are position-indexed (Position, Name, Year, URL, Description)
- All string fields are double-quoted with internal quotes escaped as double-double-quotes
- Output is streamed directly to the browser via `res.send()` with `Content-Disposition: attachment` — no intermediate disk write

2.3 Code Consolidation

Unify all scoring logic under a single shared `scoreItem(movie, taste)` function:

- The watchlist ranking endpoint (GET `/api/recommendations`) uses `scoreItem()`
- The rewatch ranking endpoint (GET `/api/rewatch-ranking`) uses `scoreItem()` as a modifier
- The export endpoint (GET `/api/export-ranked-watchlist`) uses `scoreItem()` for consistent ordering
- No separate scoring code paths — any change to the scoring formula is reflected everywhere simultaneously

2.4 Additional Fixes (In-Sprint)

Fix	Root Cause	Resolution
Overlap detection	Title+year-only matching missed rewatches with differing URL formats	<code>computeOverlapAndClean()</code> now uses URL as primary key, title+year as fallback
Rewatch ranking formula	$0.6 \times \text{rating} + 0.4 \times \text{model}$ allowed low-rated films to outrank high-rated ones	New formula: $\text{rewatchScore} = \text{userRatingNorm} \times (0.70 + 0.30 \times \text{modelScore})$
Derived cache invalidation	Old entries missing <code>ratedItemsMeta</code> caused silent neighbour failures on restart	<code>loadDerivedCacheFromDisk()</code> checks for <code>ratedItemsMeta</code> and discards invalid entries

3. Deliverables

1. Refactored Node.js/Express backend (`server.js`) with unified `scoreItem()`, triple duplicate prevention, and corrected rewatch formula
2. Vanilla JS frontend (`public/js/app.js`) and stylesheet (`public/css/main.css`)
3. Fully functional CSV export pipeline — Letterboxd v7 compliant, browser-streamed
4. Software Engineering Documentation: SRS, SDS, and Test Plan with 86 manually executed test cases
5. Electron desktop wrapper (`electron/main.js`, `preload.js`, `setup.html`) with first-run setup wizard
6. Recorded and live demonstration of the working system

4. Timeline

This is a 24-hour rapid refactoring sprint. The sprint scope is fixed — all items listed above are in-scope and non-negotiable. No new features are introduced; the sprint is exclusively bug fixes, architectural consolidation, and documentation.

Phase	Activities	Outcome
Data Layer	Overlap detection fix, triple duplicate check, cache invalidation fix	Clean data integrity across all mutation endpoints
Scoring Layer	Unified <code>scoreItem()</code> , rewatch formula correction, cache invalidation wiring	Single source of truth for all scoring

Phase	Activities	Outcome
Export Layer	v7 CSV format compliance, browser streaming, shared scoreItem() in export	Correct, importable Letterboxd export
Documentation	SRS, SDS, Test Plan authorship; 86 test case execution	Complete SE documentation package
QA Sign-Off	Manual regression pass against all 86 test cases	All cases passing, ready for deployment